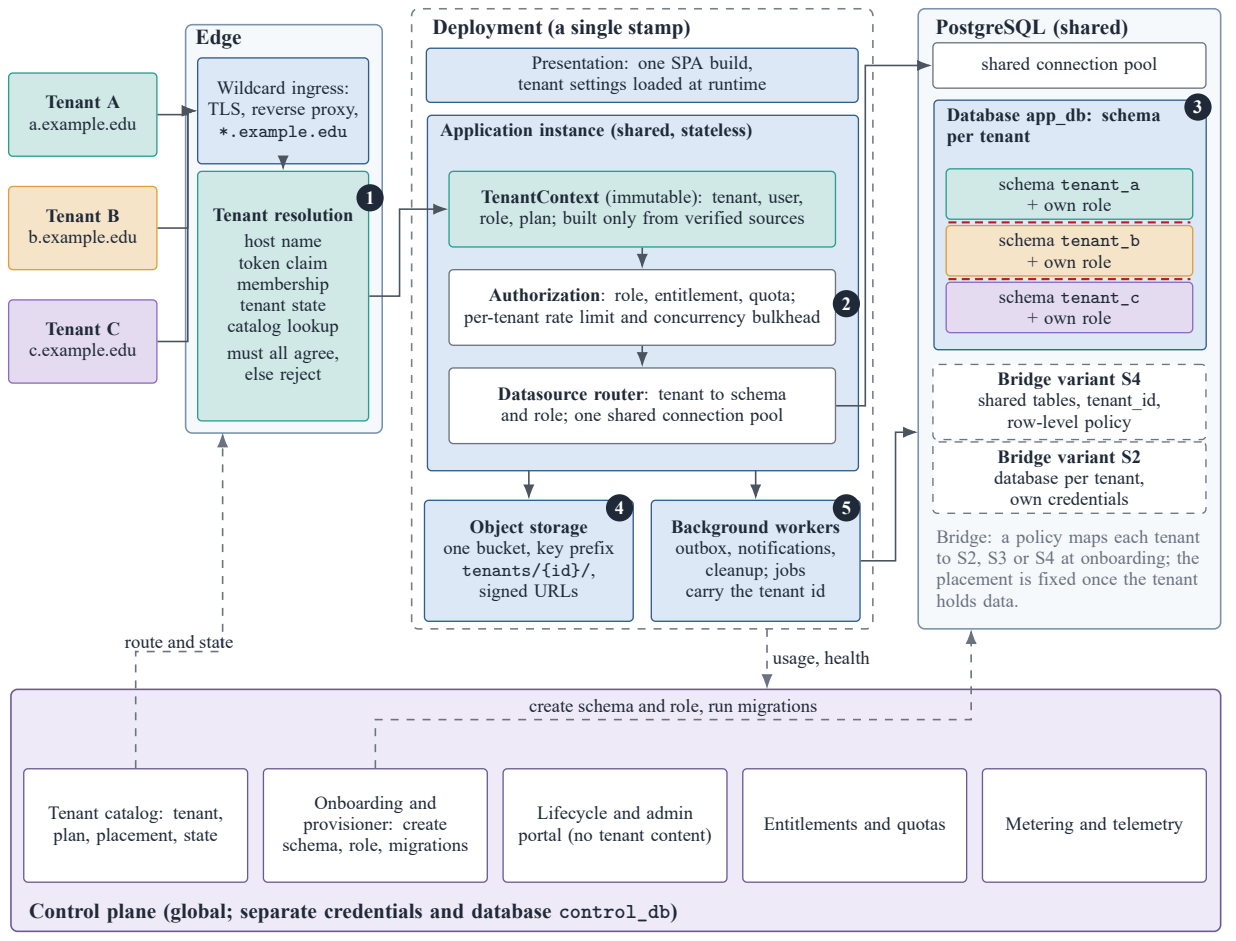


1. Edge: host name, token, membership and tenant state must agree
2. Service: tenant context only from verified sources; role, entitlement and quota checked in every use case
3. Data: per-tenant database role without privileges on other schemas, so a routing mistake fails closed
4. Objects: server-generated key prefix per tenant; access authorized before a signed URL is issued
5. Async: every job and message carries the tenant identifier; the worker restores the context and deduplicates